FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Asthma

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Mawanda

An acute asthma attack in a child is graded in the first two minutes (can the child talk or feed, SpO_2, respiratory rate, pulse, chest sounds), and treated at once with oxygen to SpO_2 94–98%, repeated salbutamol by spacer or nebuliser, early oral or IV steroid, and escalation to ipratropium, IV magnesium and then aminophylline for severe or life-threatening attacks; no child goes home without a written plan, a spacer, checked inhaler technique and a controller inhaler.

Must know before the ward round

1. Severity (UCG 2023/BTS): SEVERE = can't complete sentences or too breathless to talk/feed, SpO_2 <92%, HR >140 (2–5 y) or >125 (>5 y), RR >40 (2–5 y) or >30 (>5 y), PEF 33–50%. LIFE-THREATENING = silent chest, cyanosis, poor effort, exhaustion, drowsiness/confusion, hypotension, PEF <33%.
2. A quiet child with a silent chest and falling respiratory rate is tiring, not improving.
3. Oxygen to keep SpO_2 94–98% (GINA 2025); UCG 2023 says high-flow oxygen to keep SpO_2 ≥94%.
4. Salbutamol: pMDI 100 µg/puff via spacer 2–10 puffs (one puff at a time, 5 breaths each), or nebulised 2.5 mg (<5 y; UCG 2023) / 2.5–5 mg (≥5 y), every 20 minutes for the first hour, then space out by response.
5. Ipratropium 250 µg nebulised (children <12 y) every 20–30 min for the first 2 hours in severe/life-threatening attacks, mixed with salbutamol (UCG 2023).
6. Steroid for every attack needing more than one dose of salbutamol: prednisolone 1–2 mg/kg once daily (max 20 mg <2 y, 30 mg 2–5 y, 40 mg 6–11 y) for 3 days; or hydrocortisone 4 mg/kg IV 6-hourly (max 100 mg) if vomiting or too ill to swallow.
7. IV magnesium sulfate 40–50 mg/kg (max 2 g) over 20 minutes is the first IV bronchodilator for severe attacks not responding to inhaled therapy (GINA 2025, BTS/SIGN, RCH).
8. Aminophylline 5 mg/kg IV over 20 minutes (omit the load if on oral theophylline), then 1 mg/kg/h; UCG 2023 still lists it but discourages it because of toxicity.
9. Never sedate an asthmatic child, never give routine antibiotics (fever = think pneumonia), and never give a β-blocker.
10. Discharge when stable on salbutamol 3–4-hourly, SpO_2 ≥94% in air, PEF ≥60–80% of best, inhaler technique checked, controller started or stepped up, written action plan, review in 2 days to 1 week.
11. Chronic control (GINA 2025): 6–11 y daily low-dose ICS from Step 2; ≥12 y as-needed ICS-formoterol (AIR) is preferred; ≤5 y daily low-dose ICS if attacks are frequent or severe. A home-made 500 mL bottle spacer works as well as a commercial one.
12. Most childhood attacks are triggered by viral infections; smoke from charcoal/firewood, dust and cold are the common Ugandan triggers.

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1. Definition

Asthma is a chronic inflammatory airway disease with **variable** respiratory symptoms (wheeze, cough, breathlessness, chest tightness) and **variable expiratory airflow limitation** (GINA 2025).

Acute asthma (exacerbation, "asthma attack") is a sudden or gradual worsening of symptoms and lung function beyond the child's usual day-to-day variation that needs a change in treatment. It is caused by bronchospasm, airway wall oedema and mucus plugging in the small airways.

Term	Meaning
Wheeze	Continuous, high-pitched, mainly **expiratory** musical sound from narrowed intrathoracic airways
Stridor	Inspiratory sound from the larynx or trachea (upper airway); not asthma
Preschool wheeze	Wheeze under 5 years. **Episodic viral wheeze**: only with colds, well in between. **Multiple-trigger wheeze**: also with exercise, laughing, smoke, at night. Multiple-trigger wheeze is more likely to be asthma
Status asthmaticus	Older term for a severe attack not responding to repeated standard bronchodilator therapy; now called **acute severe** or **life-threatening asthma**
Near-fatal asthma	An attack needing ventilation, or with a rising $PaCO_2$

Examinable thresholds:

- **Severe**: SpO_2 <92% in air, or PEF 33–50% of best or predicted.
- **Life-threatening**: PEF <33%, silent chest, cyanosis, exhaustion or altered consciousness.
- **Near-fatal**: needs ventilation, or $PaCO_2$ is rising.
- **Reversibility (diagnosis, when well, age ≥5–6 years)**: FEV_1 rises ≥12% after salbutamol (UCG 2023).

Classification of the attack (UCG 2023, from BTS/SIGN): mild–moderate, severe, life-threatening. Full criteria in section 5.

2. Epidemiology

- **Global**: asthma is the commonest chronic disease of childhood. WHO estimates about 363 million people affected and about 442,000 deaths in 2023, mostly in low- and middle-income countries Verify *figures against the current WHO fact sheet*.
- **Uganda and sub-Saharan Africa**: reliable national data for Ugandan children are not available. School surveys suggest wheeze in about one in ten children, more in urban Kampala than rural areas Verify *primary source and figure*. Most are **undiagnosed, or use only salbutamol tablets or syrup**.
- **Presentation**: one of the commonest reasons a school-age child is carried into the emergency room at night. Children still die, usually because the attack was **under-graded**, treated with too little salbutamol, given oxygen late, or sent home without a plan.
- **Oral treatment culture**: many children are still treated with oral salbutamol and oral aminophylline. These work slowly, cause tremor and tachycardia, and are no substitute for an inhaler with a spacer.

Services by level of care (equipment; tests are in section 8 and drugs in section 9) Verify *compiled from the source notes and local experience, not an MoH equipment list*

Level	Usually available	Often missing
HC III	Salbutamol inhaler, sometimes a nebuliser, malaria RDT, HIV test	Oxygen, pulse oximeter, spacer, peak flow meter
HC IV / General Hospital	Oxygen concentrator, pulse oximeter, nebuliser, basic laboratory (FBC), X-ray at some HC IVs and all hospitals	Peak flow meter, commercial spacers, HDU
Regional Referral Hospital (RRH)	All the above, chest X-ray, HDU, serum electrolytes	Blood gas machine at many RRHs, PICU
National Referral Hospital (NRH, Mulago) and large PNFP hospitals (Mengo, Lubaga, Nsambya)	Blood gas, HDU, echocardiography, bronchoscopy; PICU and ventilation at Mulago and a few private hospitals	PICU beds are very few

Uganda pearl. Magnesium sulfate is on every labour ward (50% ampoules for eclampsia), so it is usually available even when the paediatric ward has none. Borrow it, but calculate the paediatric dose (40–50 mg/kg, max 2 g) and dilute it: the eclampsia loading dose (4 g IV) would be a dangerous overdose in a child.

3. Aetiology and Risk Factors

Inherited susceptibility plus environmental exposure cause allergic (Th2) airway inflammation. An attack needs a trigger on top.

Triggers (commonest in our setting first)

- **Viral infections**, especially rhinovirus colds: the commonest trigger.
- **Indoor smoke**: charcoal stoves, firewood, kerosene lamps, cigarette smoke.
- **Dust and cold**: house dust mite, dusty roads in the dry season, cold early mornings.
- **Allergens**: cockroaches, pets, moulds.
- **Exercise, laughing, strong perfumes.**
- **Drugs**: aspirin and NSAIDs (in sensitive children), β-blockers.
- **Food or sting**: think anaphylaxis, not asthma alone (section 7).
- **Poor control**: no inhaled corticosteroid (ICS), no spacer, poor adherence or technique, salbutamol tablets only, cost of inhalers.

Risk factors for asthma

Non-modifiable	Modifiable
Family history of asthma or atopy	Tobacco smoke, including in pregnancy
Eczema, allergic rhinitis, food allergy	Indoor biomass smoke (charcoal, firewood, kerosene)
Male sex in childhood	Damp housing, dust mite, cockroaches

Risk factors for a fatal attack (GINA 2025). Ask about every one; lower your threshold to escalate if any is present.

Modifiable	Non-modifiable
Not on ICS; poor adherence	Previous ICU admission or intubation
More than one salbutamol canister a month	≥2 admissions or ≥3 emergency visits in the last year
Poor inhaler technique, no spacer	Recent or current oral steroid courses
Psychosocial problems	Food allergy

Modifiable	Non-modifiable
Poor perception of severity (child or family)	

4. Pathophysiology

Trigger and injury. Chronic eosinophilic Th2 inflammation (IL-4, IL-5, IL-13, mast cells, IgE) makes the airway hyper-responsive. A trigger, usually a rhinovirus cold, then causes three lesions:

- **Bronchial smooth-muscle spasm**: reverses in minutes with a β_2-agonist; ipratropium blocks the vagal component.
- **Mucosal oedema and inflammatory cell influx**: reverses over hours with corticosteroids, which act from about 3–4 hours.
- **Mucus plugging**: clears slowly. This is why severe attacks respond slowly and can cause atelectasis.

Compensation. Narrowing is worst in expiration, when rising intrathoracic pressure compresses the airways. Air is trapped and the chest hyperinflates. Ventilation–perfusion (V/Q) mismatch lowers oxygen. The child breathes faster and harder, so early $PaCO_2$ is low.

Why children decompensate quickly. Resistance is proportional to $1/\text{radius}^4$, so a little oedema narrows a child's small airway far more than an adult's. A compliant chest wall is drawn in (recession) instead of lifting, so effort is wasted. Children have fewer collateral channels, and the diaphragm tires early. The rate falls, $PaCO_2$ rises to normal then high, and hypoxaemia worsens.

Blood gases through an attack

Phase	PaO_2 / SpO_2	$PaCO_2$	Meaning
Early	Normal or slightly low	**Low** (hyperventilation)	Expected
Worsening	Falling (V/Q mismatch)	**Normal**	**Danger**: a "normal" $PaCO_2$ in a breathless child means tiring
Life-threatening	Low	**Rising** (>6 kPa / 45 mmHg)	Respiratory failure: HDU or ICU, consider ventilation

Trigger: virus, smoke, dust

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Inflamed, hyper-responsive airway

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Injury: bronchospasm, oedema, mucus plugs

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Air trapping, V/Q mismatch

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Compensation: tachypnoea, recession, tachycardia, low PaCO2

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Respiratory muscle fatigue

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Decompensation: falling RR, silent chest, rising PaCO2

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Drowsiness, hypotension, bradycardia, arrest

Feature links

- Narrowed airways, turbulent flow → wheeze; worse narrowing → wheeze in both phases.
- Near-zero airflow → silent chest.
- Inflammation and mucus → cough, worse at night.
- Expiratory collapse → air trapping → hyperinflation, hyper-resonance.
- Raised resistance → work of breathing → tachypnoea, recession, nasal flaring, accessory muscles, tripod posture, cannot talk or feed.
- V/Q mismatch → hypoxaemia → low SpO_2, agitation, cyanosis.
- Hypoxaemia and sympathetic drive → tachycardia.
- Muscle fatigue → falling RR, poor effort → rising $PaCO_2$ → drowsiness, confusion.
- Hyperinflation → reduced venous return; plus hypoxic myocardium → hypotension, bradycardia.
- Alveolar rupture → pneumothorax, pneumomediastinum, neck crepitus.
- Mucus plug → collapse → focal reduced air entry.
- Large negative intrathoracic pressure → pulmonary oedema if IV fluid is excessive.
- Salbutamol → tremor, tachycardia, low potassium, high lactate, high glucose.

Why the drugs work (and their side effects)

Drug	Mechanism	Main side effects
Salbutamol	β_2-agonist: raises cAMP, relaxes airway smooth muscle in minutes	Tachycardia, tremor, **hypokalaemia**, raised **lactate**, hyperglycaemia, restlessness
Ipratropium	Antimuscarinic: blocks vagal bronchoconstriction	Dry mouth; adds to salbutamol in the first hours only
Corticosteroids	Switch off inflammatory gene transcription, reduce oedema, restore β-receptor response	Effect from **3–4 h**; hyperglycaemia, mood change; short courses are safe
Magnesium sulfate	Blocks calcium channels in smooth muscle → bronchodilation	Flushing, **hypotension**, weakness, loss of reflexes at high levels
Aminophylline	Phosphodiesterase inhibitor, adenosine antagonist	**Narrow therapeutic window**: vomiting, tachyarrhythmia, **seizures**
Oxygen	Corrects V/Q-mismatch hypoxaemia	None in asthma at these doses (unlike COPD)

5. Classification / Staging

Severity of the acute attack (UCG 2023, adapted from BTS/SIGN and GINA). Any **one** feature of a higher grade puts the child in that grade; not all features need to be present.

Feature	Mild–moderate	Severe	Life-threatening
Talking / feeding	Talks in sentences; feeds	**Cannot complete sentences in one breath**; too breathless to talk or feed	Unable to talk; exhausted
Behaviour	Alert, may be agitated	Agitated, sits hunched forward	**Drowsy, confused**, floppy
SpO_2 in air	**≥92%**	**<92%**	<92%, often much lower; **cyanosis**
Respiratory rate	≤40 (2–5 y); ≤30 (>5 y)	**>40 (2–5 y); >30 (>5 y)**	May be **falling** (exhaustion); poor effort
Pulse	≤140 (2–5 y); ≤125 (>5 y)	**>140 (2–5 y); >125 (>5 y)**	**Bradycardia**, arrhythmia, **hypotension**

Feature	Mild–moderate	Severe	Life-threatening
Accessory muscles	Not used, or mild	**Used**: sternomastoid, subcostal and intercostal recession, nasal flaring	Exhausted; paradoxical breathing
Chest sounds	Wheeze	Loud wheeze, both phases	**Silent chest**
PEF (≥5–6 y who can do it)	**>50%** of predicted or best	**33–50%**	**<33%**

If the parent or child says this is the worst attack ever, believe them.

GINA 2025 uses "mild or moderate" and "severe", with slightly different cut-offs for 6–11 years (for example pulse >120/min, SpO_2 <90%, PEF ≤50%) Verify *GINA 2025 thresholds*. Use the UCG table in Ugandan practice and exams.

Control of the underlying asthma (GINA 2025; used for the discharge plan)

Level of control (last 4 weeks)	Criteria
Well controlled	Daytime symptoms ≤2/week (≤1/week for ≤5 y), no night waking, reliever ≤2/week, no activity limitation
Partly controlled	1–2 of these not met
Uncontrolled	3–4 not met

UCG 2023 still uses the older chronic-severity steps: **intermittent, mild persistent, moderate persistent, severe persistent** (section 9, Chronic control). Know both vocabularies.

6. Clinical Features

History

Take it **while the first salbutamol is running**. Each question has a reason.

- **Symptoms**: breathlessness, wheeze, cough, chest tightness; in young children, poor feeding.
- **Age** → under 2 years: bronchiolitis or viral wheeze more likely; a first-ever wheeze in an infant is usually bronchiolitis. ≥5–6 years: can do PEF.
- **Is this asthma?** Previous wheeze, night cough, exercise cough, relief with inhalers, atopy (eczema, allergic rhinitis), family history.
- **Onset and trigger** → sudden choking while eating or playing = foreign body; food, sting or drug with rash or lip swelling = anaphylaxis; cold = viral exacerbation.
- **Treatment already given**: number of puffs, spacer or not, nebuliser, steroid, and the response. A child who had 10 puffs at home and is still severe is a **poor responder**. **Salbutamol tablets do not count**: a child "on treatment" may never have had an inhaled drug.
- **Danger history** (fatal-attack risk factors, section 3): previous ICU or intubation, admissions or emergency visits in the past year, oral steroid courses, salbutamol canisters per month → lower your threshold to escalate.
- **Usual treatment and adherence**: is there an inhaled steroid at home? Is it taken? Who gives it? Is there a spacer? → informs the discharge plan.
- **Fever, chest pain, purulent sputum** → pneumonia. **Fever is not part of asthma**: viral triggers cause low-grade fever, but a high fever with focal signs means look for pneumonia and treat it (UCG 2023 directs you to the pneumonia guideline when fever is present) Verify *UCG wording*.

- **TB contact, weight loss, cough >2 weeks** → TB can cause wheeze by lymph-node compression of a bronchus.
- **HIV status, cotrimoxazole, ART** → PCP, lymphoid interstitial pneumonitis (LIP) and TB in the differential; drug interactions (section 13).
- **Immunisation (UNEPI, including PCV and measles–rubella) and nutrition** → measles and pertussis; malnutrition changes fluids and risks.
- **Home environment**: charcoal or firewood indoors, kerosene lamps, smokers, dust, school, cost of inhalers → trigger control and plan.

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: be **opportunistic** (observe first, auscultate while the child is quiet, leave distressing parts such as the throat and ears to last), use play and distraction, and judge every vital sign against **age-specific normal values**. In an emergency the ETAT and ABCDE assessment (section 9) comes first; complete the full sequence once salbutamol is running.

Introduction, consent and preparation

- **Introduction and identity**: introduce yourself; confirm the child's name and age and the parent's or carer's relationship to the child.
- **Explanation and consent**: explain the examination; gain consent from the parent and assent from the child.
- **Hand hygiene**: alcohol rub or soap and water before touching the child.
- **Position and exposure**: the child sits up on the parent's lap (never lie a breathless child flat); expose the chest as needed in a warm room; keep a toy or phone light for distraction.

General inspection (end of the bed)

Sign	Mechanism	What it means
Well or unwell; cannot complete sentences or feed; tripod posture	High work of breathing leaves no breath for speech or sucking	Severe attack
Agitation	Hypoxaemia	Hypoxaemia until proven otherwise: give oxygen, never a sedative
Drowsiness, confusion, floppiness	Hypoxaemia and hypercapnia	Life-threatening
Central cyanosis	Severe V/Q mismatch	Life-threatening
Audible wheeze or stridor	Intrathoracic versus extrathoracic narrowing	Wheeze: asthma; stridor: upper airway, not asthma
Sweating, dry mouth, sunken eyes	Work of breathing, poor intake	Dehydration: oral or maintenance fluids
Wasting, oedema	Malnutrition	SAM changes fluids and potassium (section 13)
Dysmorphism, cerebral palsy	Neuromuscular or syndromic disease	Recurrent aspiration as a mimic

Vital signs and growth: count RR for a **full minute**; HR; SpO_2 in air if safe, then on oxygen; temperature; BP if severe; capillary refill; weight and height. Compare RR and HR with the age cut-offs in section 5 (RR >40 at 2–5 y or >30 over 5 y; HR >140 at 2–5 y or >125 over 5 y = severe). A falling RR without improvement means exhaustion; tachycardia may be salbutamol; bradycardia and hypotension are pre-arrest (hyperinflation reduces venous return; hypoxic myocardium).

Hands, face and neck

Sign	Mechanism	What it means
Clubbing	Chronic suppurative or interstitial lung disease (asthma does not cause it)	Not asthma: bronchiectasis, cystic fibrosis, TB, HIV lung disease
Peripheral cyanosis, cool hands, slow capillary refill	Poor perfusion	Late shock
Fine tremor, fast pulse	β_2-agonist effect	Salbutamol effect, not necessarily a worse attack
Central cyanosis of the tongue	Hypoxaemia	Life-threatening
Allergic shiners, eczema on the face	Atopy	Supports asthma
Nasal flaring	Increased work of breathing	Respiratory distress
Nasal polyps	Chronic sinonasal disease	Think cystic fibrosis
Lip or facial swelling, urticaria	Mast-cell degranulation	Anaphylaxis: IM adrenaline first
Parotid swelling	Lymphocytic infiltration in HIV	LIP
Tracheal tug, sternomastoid use	Accessory muscles recruited	High work of breathing
Neck crepitus (subcutaneous emphysema)	Alveolar rupture, air leak	Pneumomediastinum or pneumothorax
Cervical lymph nodes	Reactive or TB	TB in the differential

Respiratory system

Step	Sign	Mechanism	What it means
Inspection	Hyperinflated (barrel) chest	Expiratory collapse traps air	Air trapping
Inspection	Subcostal, intercostal and suprasternal recession; abdominal breathing	Compliant chest wall drawn in against high resistance	High work of breathing
Inspection	Harrison sulci	Chronic diaphragmatic pull	Long-standing poorly controlled asthma
Inspection	Asymmetrical movement	Pneumothorax, collapse, foreign body	Not asthma alone
Palpation	Trachea deviated	Tension pneumothorax (away) or collapse (towards)	Urgent chest X-ray; decompress if tension
Palpation	Reduced expansion, one side	Pneumothorax, collapse, consolidation	Focal pathology
Percussion	Hyper-resonant both sides	Air trapping	Expected in a severe attack
Percussion	Hyper-resonant one side	Pneumothorax	Air leak
Percussion	Dull	Consolidation or collapse	Pneumonia or mucus plug
Auscultation	Expiratory wheeze	Turbulent flow through narrowed airways	Asthma
Auscultation	Wheeze in both phases	Narrowing in inspiration as well	Severe narrowing

Step	Sign	Mechanism	What it means
Auscultation	Silent chest	Almost no airflow	Life-threatening
Auscultation	Unequal air entry	Pneumothorax, foreign body, mucus plug	Not asthma alone; chest X-ray
Auscultation	Focal crackles or bronchial breathing with fever	Consolidation	Pneumonia
Auscultation	Fine crackles with wheeze under 2 years	Bronchiolar inflammation	Bronchiolitis

Auscultate early if the child is quiet on the parent's lap; then return to the rest of the sequence.

Completing the examination

- **Cardiovascular**: murmur or gallop (heart failure mimicking asthma).
- **Abdomen**: liver edge. A liver pushed down by flattened, hyperinflated diaphragms is soft with a normal span; a large, tender liver with a murmur suggests heart failure.
- **Skin**: eczema, urticaria, BCG scar.
- **Ears, nose and throat, last**: nasal discharge, tonsils, ear drums.
- **Bedside tests**: SpO_2 (if not already done), PEF at ≥5–6 years, blood glucose. **Plot growth**: weight-for-age and height-for-age Z-scores (poor growth: uncontrolled asthma, repeated steroids, TB, HIV).
- **Thank** the parent and child, re-dress the child, **wash hands**.
- **Summarise findings** with a severity grade and an immediate plan (model presentation in section 8).

Document these positives: talk/feed ability and AVPU; SpO_2 in air and on oxygen; RR and HR; recession and accessory muscle use; response after each salbutamol dose, time-stamped.

Document these negatives: "no silent chest; air entry equal"; "no focal crackles, no bronchial breathing"; "no stridor; no urticaria or angioedema"; "trachea central; no surgical emphysema".

Red flags

- **Life-threatening asthma, act now and call the most senior person**: silent chest, cyanosis, SpO_2 <92% despite oxygen, poor respiratory effort, exhaustion, **drowsiness or confusion**, bradycardia or hypotension, PEF <33%, a normal or rising $PaCO_2$.
- **No better after one hour** of 20-minutely salbutamol.
- **A quiet chest is not a better chest.** A child who stops wheezing, becomes drowsy and breathes more slowly while SpO_2 falls is about to arrest. Do not be reassured by a falling RR unless the child looks better and SpO_2 is rising.
- **Not asthma, or asthma plus something**: unilateral wheeze or reduced air entry (foreign body, pneumothorax); fever with focal crackles (pneumonia); stridor (upper airway); urticaria, facial swelling and hypotension (**anaphylaxis: IM adrenaline first**, section 9); murmur, hepatomegaly and crackles (heart failure); chronic cough, weight loss and a TB contact. "All that wheezes is not asthma."

Thinking like a clinician

- **Grade, then treat, then re-grade every 20–30 minutes.** The grade changes, and so must the plan.
- **"All that wheezes is not asthma."** Under 2 years, think bronchiolitis first. Sudden onset with choking = foreign body until proven otherwise. **Unilateral** findings are never asthma alone.
- **Fever is not part of asthma.** UCG 2023: "if fever present, refer to pneumonia" Verify *UCG wording*. Viral triggers cause low-grade fever, but a high fever with focal signs means look for pneumonia and treat it.
- **Don't be reassured by a falling respiratory rate** unless the child also looks better and SpO_2 is rising.

- **Salbutamol tablets at home do not count as treatment.** A child "on treatment" may never have had an inhaled drug.
- **Agitation is hypoxia until proven otherwise**: give oxygen, never diazepam or promethazine.
- **Anaphylaxis can look like asthma.** Wheeze + urticaria/angioedema + hypotension after a food, drug or sting → **IM adrenaline 0.01 mg/kg (0.01 mL/kg of 1:1000), max 0.5 mg**, into the outer thigh, before anything else.
- **Every admission is a failure of prevention.** Ask yourself: why did this child get this bad? (No ICS? No spacer? Charcoal stove? Cost?)

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Bronchiolitis	Under 2 years (peak 2–6 months), first episode, coryza then cough, **fine crackles** with wheeze, little response to salbutamol	Clinical; SpO_2
Pneumonia	Fever, fast breathing, chest indrawing, **focal crackles or bronchial breathing**, no response to bronchodilators	Chest X-ray, FBC
Foreign body aspiration	Sudden choking while eating or playing (groundnuts, beans, maize), **unilateral** wheeze or reduced air entry, no atopy	Inspiratory and expiratory chest X-ray; bronchoscopy
Anaphylaxis	Food, drug or sting; urticaria, angioedema, hypotension, vomiting	Clinical; responds to IM adrenaline
Croup / upper airway obstruction	**Inspiratory stridor**, barking cough, hoarse voice	Clinical
Heart failure (VSD, rheumatic heart disease, cardiomyopathy)	Murmur, gallop, **hepatomegaly**, basal crackles, poor growth, sweating with feeds	Chest X-ray (cardiomegaly), echocardiography
Pulmonary TB, TB lymph nodes	Cough >2 weeks, weight loss, fever, TB contact; wheeze from node compression	GeneXpert, chest X-ray (hilar nodes), tuberculin skin test
HIV-related lung disease (LIP, PCP, bronchiectasis)	HIV-positive, clubbing, parotid swelling (LIP), hypoxia out of proportion (PCP)	HIV test, chest X-ray
Recurrent aspiration, gastro-oesophageal reflux	Cough with feeds, cerebral palsy, vomiting	Clinical; swallow assessment
Bronchiectasis, cystic fibrosis	Chronic wet cough, **clubbing**, failure to thrive, nasal polyps (CF)	Chest X-ray; sweat test (rarely available)
Loeffler syndrome (ascaris larval migration), tropical pulmonary eosinophilia	Wheeze with worms	FBC eosinophilia, stool microscopy

8. Clinical Workup

Diagnosis and grade are clinical. Treat first; investigate selectively. "Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list.*

Bedside

Test	When / how	Expected finding	How it changes management	Available at
Pulse oximetry	Every child; correct-size probe, warm finger or toe, wait for a good waveform	**<92% in air = severe**; may overestimate SpO_2 in dark skin, so read with the clinical picture	Grades the attack; titrates oxygen to 94–98%	HC IV and above; often missing at HC III
Peak expiratory flow	≥5–6 years; best of 3 blows standing, compare with personal best or predicted for height; repeat 15–30 min after each treatment	<50% severe; <33% life-threatening; >60–80% for discharge	Objective grade and response	Often missing even at HC IV; RRH and NRH
Response to salbutamol	Re-grade 15–20 min after each dose	Better talking, lower RR and work of breathing, rising SpO_2	The most important "test"; a good response supports asthma	All levels
Random blood glucose	Poor intake; prolonged treatment	Low if not fed; high after salbutamol or steroid	Treat hypoglycaemia	HC IV and above; variable at HC III
Malaria RDT	Fever	Co-existing malaria is common; fever needs a cause	Antimalarial	All levels
Weight	Every child	If no scale: (age + 4) × 2 kg for 1–10 years	Every dose is per kg	All levels

Basic lab

Test	When	Expected finding	How it changes management	Available at
Serum potassium	Repeated or continuous salbutamol; aminophylline	**Hypokalaemia** (<3.5 mmol/L)	Add potassium to IV fluids	RRH, NRH
FBC	Fever, suspected infection	Neutrophilia (pneumonia; but steroids also raise the white cell count); eosinophilia (atopy, worms)	Antibiotics only for clear bacterial infection	HC IV and above
HIV test	Status unknown	Positive	ART, cotrimoxazole; widen differential (LIP, PCP, TB)	All levels

Imaging

Test	When	Expected finding	How it changes management	Available at
Chest X-ray	**Not routine.** Only if life-threatening, poor response, suspected pneumothorax, focal signs, fever, first episode with doubt about the diagnosis, or foreign body	Hyperinflation and peribronchial thickening (expected in asthma); look for **pneumothorax, pneumomediastinum, collapse or consolidation, foreign body**	Chest drain for tension or large pneumothorax; antibiotics for pneumonia; bronchoscopy for foreign body	Some HC IVs; all hospitals

Specialised / referral-level

Test	When	Expected finding	How it changes management	Available at
Blood gas (capillary or arterial)	Life-threatening or not responding; HDU	$PaCO_2$ **>6 kPa (45 mmHg)** with hypoxaemia = respiratory failure; lactic acidosis from salbutamol	HDU or ICU; consider ventilation	Mulago and private ICUs; many RRHs have no gas machine
Theophylline level	On aminophylline with vomiting, arrhythmia or seizures	Toxic level	Stop the infusion	Rarely available: watch clinically
GeneXpert (sputum, gastric aspirate or stool)	Cough >2 weeks, weight loss, TB contact	MTB detected	TB treatment (childhood TB topic)	HC IV and hospitals; samples referred from lower levels
Spirometry with reversibility	When well, ≥5–6 years, in clinic	FEV_1/FVC <0.8–0.9; **FEV_1 rise ≥12%** after salbutamol confirms asthma (UCG 2023)	Confirms the diagnosis	NRH, some RRHs and private clinics
Echocardiography	Murmur, heart failure	Structural heart disease	Cardiac treatment	Some RRHs; NRH
Bronchoscopy	Suspected foreign body	Foreign body seen and removed	Definitive treatment	NRH

Putting it together

Model one-line summary

> "A 7-year-old boy weighing 22 kg (weight-for-age Z-score −0.5), fully immunised, HIV-negative, known asthmatic on a salbutamol inhaler only, with a 1-day history of worsening breathlessness after a cold. On arrival he could speak only in words, SpO_2 89% in air, RR 38, HR 138, widespread wheeze, PEF 40% of predicted: **acute severe asthma** triggered by a viral upper respiratory tract infection, in a child with poorly controlled asthma on no controller."

Problem list

1. Acute severe asthma, on treatment, response being assessed.
2. Hypoxaemia (SpO_2 89% in air).
3. Poorly controlled chronic asthma: no ICS, no spacer, charcoal stove at home.
4. Viral URTI trigger (no evidence of pneumonia).

Working diagnosis: acute severe exacerbation of asthma, viral trigger, uncontrolled background asthma.

Model case presentation (ward round)

> "Brian is a 7-year-old boy from Kawempe, weighing 22 kg (weight-for-age Z-score −0.5, height-for-age Z-score 0), fully immunised by UNEPI including PCV and both measles–rubella doses, HIV-negative (tested last year), on a normal family diet. He is a known asthmatic diagnosed at 4 years, using only a salbutamol inhaler without a spacer; no inhaled steroid. He presented at 2 a.m. with one day of cough and worsening breathlessness after a cold; his mother cooks on a charcoal stove indoors. He has had two admissions this year, no ICU stay. On arrival he could say only words, was sitting forward, SpO_2 89% in air, RR 38, HR 138, temperature 37.6 °C, with intercostal recession and widespread expiratory wheeze, equal air entry, no crackles; PEF 40% of predicted. Assessment: **acute severe asthma**, viral trigger, on a background of uncontrolled asthma. He received oxygen, three back-to-back nebulisers of salbutamol 5 mg with

> ipratropium 250 µg, and prednisolone 40 mg. At 1 hour he speaks in sentences, SpO_2 95% on 2 L/min, RR 28, PEF 65%. Plan: salbutamol 2-hourly then 4-hourly by spacer, prednisolone for 3 days, wean oxygen, start beclomethasone 100 µg twice daily via a bottle spacer, teach technique, written action plan, smoke advice, review in clinic in 1 week."

Worked example 1: re-grading after treatment. At 0 min: SpO_2 89%, RR 38, HR 138, speaks in words, PEF 40% → **severe**. Given oxygen, salbutamol 5 mg + ipratropium 250 µg nebulised three times over 1 hour, and prednisolone 40 mg (2 mg/kg would be 44 mg; the maximum is 40 mg). At 60 min: SpO_2 95% on 2 L/min, RR 28, HR 150, speaks in sentences, PEF 65%, tremor. **Interpretation**: responding well; the higher HR and the tremor are salbutamol effects. **Plan**: space salbutamol to 1–2-hourly, then 4-hourly; wean oxygen; continue prednisolone for 3 days; start ICS before discharge.

Worked example 2: a "normal" blood gas that is not normal. A 9-year-old girl after 2 hours of back-to-back nebulisers is drowsy, RR 22, quiet chest, SpO_2 91% on a mask. Capillary gas: pH 7.30, **$PaCO_2$ 6.4 kPa (48 mmHg)**, lactate 3.5 mmol/L. **Interpretation**: a breathless asthmatic should be blowing off CO_2 (low $PaCO_2$). A normal or high $PaCO_2$ with drowsiness and a falling RR means **exhaustion and respiratory failure**. This is life-threatening asthma: IV magnesium now (if not given), consider aminophylline or IV salbutamol in HDU, and call the anaesthetist for possible intubation.

9. Management

Immediate / emergency

First 5 minutes

1. **ETAT triage**: severe respiratory distress and central cyanosis are emergency signs. Move the child to the **resuscitation area** and sit them up on the parent's lap.
2. **Oxygen** by nasal prongs (1–2 L/min) or face mask (5–10 L/min; UCG: at least 5 L/min) to keep **SpO_2 94–98%**, and the **first salbutamol** (algorithm step 3) at once.
3. **Grade the attack** (section 5) in under 2 minutes; if severe, call for senior help and site IV access.
4. **Weigh the child** (or estimate (age + 4) × 2 kg for 1–10 years) so every later drug is weight-based.

ABCDE

- **A**: sit up. Check for stridor (upper airway, not asthma).
- **B**: oxygen and salbutamol as above; if severe, add ipratropium and drive the nebuliser with oxygen.
- **C**: HR, capillary refill, BP if severe (hypotension is late and ominous). IV access if severe. IV fluids at maintenance only if not drinking; no boluses unless shocked.
- **D**: AVPU and glucose. **Never sedate** (diazepam, promethazine, phenobarbital, chloral): restlessness is hypoxia, and sedation removes the drive that is keeping the child alive.
- **E**: temperature; urticaria or angioedema means anaphylaxis: **IM adrenaline 0.01 mg/kg (0.01 mL/kg of 1:1000), max 0.5 mg**, into the outer thigh, before anything else.

Life-threatening (drowsy, cyanosed or silent chest): back-to-back nebulised salbutamol + ipratropium on oxygen, IV access, **IV hydrocortisone**, **IV magnesium** within the first hour, call the most senior clinician and the anaesthetist, and prepare a bag-valve-mask. If the child stops breathing, start paediatric life support.

Emergency algorithm (UCG 2023, GINA 2025, BTS/SIGN)

1. **Assess and grade** (2 minutes). Weigh.
2. **All grades**: sit up; oxygen if SpO_2 <94% (target 94–98%).
3. **Salbutamol**:

- **Mild–moderate**: pMDI 100 µg/puff via spacer, **2–10 puffs** (one puff at a time, 5 tidal breaths per puff), **every 20 minutes for 1 hour**; or nebulised 2.5 mg (<5 y) / 2.5–5 mg (≥5 y).
- **Severe or life-threatening**: **nebulised salbutamol 2.5–5 mg + ipratropium 250 µg**, oxygen-driven (6–8 L/min), every 20 minutes (back-to-back if life-threatening).

4. **Steroid in the first hour** for any attack needing more than one salbutamol dose: prednisolone 1–2 mg/kg orally (max 20 mg <2 y, 30 mg 2–5 y, 40 mg ≥6 y), or **hydrocortisone 4 mg/kg IV** (max 100 mg) 6-hourly if vomiting, drowsy or life-threatening.
5. **Re-grade at 1 hour**:
   - **Good response** (talking, SpO_2 ≥94% in air, PEF >60–80%): space salbutamol to 1–4-hourly; observe 1–4 hours; plan discharge.
   - **Incomplete response**: admit; salbutamol 1–2-hourly + ipratropium for the first 2 hours (then 4–6-hourly); steroid; oxygen.
   - **Poor response or life-threatening**: **IV magnesium sulfate 40–50 mg/kg (max 2 g) over 20 minutes**; senior review; HDU.
6. **Still not improving**: **aminophylline** 5 mg/kg IV over 20 minutes (omit the load if on oral theophylline), then 1 mg/kg/h with cardiac monitoring; or **IV salbutamol** in HDU (15 µg/kg over 10 minutes, then 1–5 µg/kg/min) where available.
7. **Respiratory failure** (exhaustion, rising $PaCO_2$, falling consciousness): anaesthetist or ICU; bag-valve-mask; intubation by the most experienced person (ketamine induction).
8. **Throughout**: SpO_2, RR, HR and work of breathing every 15–30 minutes; potassium and glucose if treatment is prolonged.

Definitive

Principles

1. Treat by grade and **re-grade every 20–30 minutes**. The grade changes, and so must the plan.
2. **Inhaled β_2-agonist is the backbone.** A spacer is as good as a nebuliser in mild–moderate and most severe attacks, and it teaches technique.
3. **Give a steroid early** (within the first hour) for any attack needing more than a single salbutamol dose.
4. **Escalate stepwise and quickly** when there is no response: ipratropium → IV magnesium → aminophylline (or IV salbutamol in HDU) → ventilation.
5. **No sedatives, no routine antibiotics, no cough syrups, no mucolytics, no β-blockers, no chest physiotherapy** in the acute attack.
6. **Before discharge**: controller, technique, spacer, action plan, follow-up.

Non-pharmacological: upright, calm, parent present; oral fluids; remove smoke exposure; daily spacer teaching.

Drug table

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Oxygen	Nasal prongs 1–2 L/min (infants 0.5–1 L/min); face mask 5–10 L/min	Inhaled	Continuous	Until SpO_2 ≥94% in air	Target 94–98%; drive the nebuliser at 6–8 L/min in severe attacks

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Salbutamol pMDI (100 µg/puff)	2–10 puffs (GINA 6–11 y: 4–10 puffs); adult 4–10 puffs	Spacer, plus face mask if <3–4 y	Every 20 min for 1 h, then 1–4-hourly by response	Wean to 4-hourly before discharge	One puff at a time, 5 breaths each; as effective as a nebuliser; tremor, low K^+
Salbutamol nebulised	2.5 mg (<5 y); 2.5–5 mg (≥5 y), in 2–4 mL 0.9% saline; adult 5 mg	Nebulised	Every 20 min ×3, then hourly to 4-hourly; back-to-back if life-threatening	Until spacing to 4-hourly	Oxygen-driven if severe
Ipratropium bromide	250 µg (<12 y); 500 µg (≥12 y, adult)	Nebulised with salbutamol	Every 20–30 min for the first 2 h, then 4–6-hourly	Stop after 24–48 h	Severe and life-threatening attacks only (UCG 2023)
Prednisolone	1–2 mg/kg; max 20 mg (<2 y), 30 mg (2–5 y), 40 mg (6–11 y); adult 40–50 mg	Oral	Once daily	3 days in children (UCG 2023); up to 5 days if slow	No taper for courses ≤14 days; crush and mix with milk or juice
Hydrocortisone	4 mg/kg, max 100 mg/dose; adult 100 mg	IV	6-hourly	Until oral prednisolone tolerated	Vomiting, drowsy or life-threatening
Dexamethasone	0.3–0.6 mg/kg, max 16 mg	Oral or IV	Once daily	1–2 days	Alternative used in some units for vomiting children; not in UCG for asthma
Magnesium sulfate	40–50 mg/kg (0.1 mL/kg of 50% = 50 mg/kg), max 2 g; adult 1.2–2 g	IV, diluted in 0.9% saline to ≤20%	Over 20 min	Single dose	Monitor BP and reflexes; 50% is hyperosmolar and irritant; never the eclampsia dose
Aminophylline	Load 5 mg/kg (max 500 mg) over 20 min, then 1 mg/kg/h; adult infusion 0.5 mg/kg/h. WHO Pocket Book alternative: 5 mg/kg over 20 min 6-hourly	IV	Infusion	Stop when improving	**Omit the load if on oral theophylline or aminophylline**; cardiac monitor; stop if vomiting, arrhythmia or seizure
Salbutamol IV	15 µg/kg (max 250 µg) over 10 min, then 1–5 µg/kg/min	IV	Infusion	HDU only	Monitor K^+, lactate, ECG
Adrenaline	0.01 mg/kg (0.01 mL/kg of 1:1000), max 0.5 mg; adult 0.5 mg	IM, outer thigh (SC for wheeze only when no salbutamol, WHO Pocket Book)	Repeat after 5–15 min	As needed	First-line for anaphylaxis, not routine asthma
Paracetamol	15 mg/kg, max 60 mg/kg/day; adult 1 g	Oral	6-hourly	While febrile	Avoid ibuprofen and aspirin if NSAID-sensitive

Adult doses are from GINA 2025 and BTS/SIGN Verify *against UCG 2023*. WHO Pocket Book aminophylline maintenance and SC adrenaline maximum Verify *against the 2013 Pocket Book text*.

Divergence from UCG 2023

- **Prednisolone**: UCG 1 mg/kg; GINA and BTS 1–2 mg/kg with age caps Verify *current UCG dose*. UCG and international guides agree on hydrocortisone 4 mg/kg (max 100 mg) and ipratropium 250 µg under 12 years.
- **Magnesium**: UCG lists no IV magnesium for children Verify. GINA, BTS/SIGN and RCH give it before aminophylline Verify *GINA 2025 wording*. Kampala referral wards (Mulago, Mengo, Nsambya) are reported to use IV magnesium before aminophylline Verify *current practice*. Borrow 50% ampoules from the labour ward, but calculate and dilute.
- **Aminophylline**: UCG lists it (5 mg/kg slow IV) but discourages it because of poor safety; GINA does not recommend it.
- **Oxygen**: UCG ≥94%; GINA 94–98%.
- **Exams**: quote UCG in the written test, and the international sequence when asked "what else could you give?"
- **EMHSLU**: level-of-care listing for ipratropium, magnesium and ICS-formoterol not checked Verify.

Drug availability by level Verify *compiled from the source notes; check against EMHSLU level codes*

Drug	HC III	HC IV / General Hospital	RRH	NRH
Oxygen	Often missing	Concentrator	Yes	Yes
Salbutamol pMDI	Yes	Yes	Yes	Yes
Salbutamol nebules	Sometimes (if a nebuliser)	Yes	Yes	Yes
Salbutamol tablets or syrup	Yes (last resort only)	Yes	Yes	Yes
Ipratropium	No	Often missing	Yes	Yes
Prednisolone tablets	Yes	Yes	Yes	Yes
Hydrocortisone IV	Variable	Yes	Yes	Yes
Dexamethasone	Variable	Variable	Yes	Yes
Magnesium sulfate	Maternity stock; refer rather than give	Borrow from maternity	Yes	Yes
Aminophylline IV	Variable	Yes	Yes	Yes
Salbutamol IV	No	No	Some HDUs	Yes
Adrenaline	Yes	Yes	Yes	Yes
Beclomethasone inhaler	Variable	Yes	Yes	Yes
Budesonide-formoterol	No	No	Some clinics	Some clinics; private pharmacies (expensive)

Referral criteria

Level	What to do	Refer up when
Home / VHT	Salbutamol 2–4 puffs via spacer, repeat every 20 min up to 10 puffs (action plan)	Not better within 1 hour, or any danger sign: go to the nearest facility
HC III (UCG level for mild–moderate)	Salbutamol by spacer or nebuliser every 20–30 min; prednisolone 1 mg/kg; observe 30–60 min	Any severe feature, no improvement in 1 hour, relapse within 3–4 hours → HC IV

Level	What to do	Refer up when
HC IV / General Hospital (UCG level for severe)	Admit; oxygen; salbutamol + ipratropium; prednisolone or hydrocortisone; aminophylline (UCG) or magnesium sulfate (borrowed from maternity) if not responding	Oxygen or IV therapy cannot be sustained; no improvement on salbutamol, ipratropium and steroid; life-threatening asthma after first aid → RRH
RRH	All the above; magnesium; HDU monitoring; chest X-ray; IV salbutamol where available	Ventilation or PICU needed; bronchoscopy for a foreign body; blood gas unavailable and child tiring → NRH
NRH (Mulago) and a few private hospitals	Blood gas, PICU, ventilation, bronchoscopy	n/a

Level split adapted from UCG 2023 levels of care Verify.

Referral checklist: oxygen during transfer (cylinder, or concentrator with inverter); a health worker who can give salbutamol on the way; steroid given; referral note with times and doses; phone call ahead.

Monitoring and follow-up

- **Observations**: RR, HR, SpO_2, work of breathing and ability to talk or feed every 15–30 minutes initially, then 1–4-hourly as the child improves.
- **Re-grade** 15–20 minutes after every bronchodilator dose; record the time and response.
- **Potassium and glucose** if salbutamol has been continuous or aminophylline is running.
- **Follow-up** within **2 days to 1 week** of discharge (UCG: review in 48 hours for mild cases; 1–2 weeks for the chronic plan) Verify *UCG review intervals*, then at 1–3 months.

Ward care

- **Salbutamol** weaned stepwise: 20-minutely → hourly → 2-hourly → 4-hourly. Switch nebuliser to spacer as soon as possible; it trains technique.
- **Oxygen** weaned when SpO_2 ≥94% in air.
- **Fluids**: encourage oral fluids. If IV fluid is needed, **maintenance only** (Holliday–Segar: 100 mL/kg/day for the first 10 kg, 50 mL/kg/day for the next 10 kg, 20 mL/kg/day thereafter). Avoid overload: high negative intrathoracic pressure promotes pulmonary oedema, and SIADH can occur.
- **Antibiotics only** if fever with focal signs, consolidation on chest X-ray, or clear bacterial infection.
- **Steroid**: switch IV hydrocortisone to oral prednisolone as soon as the child can swallow.
- **Teach** inhaler and spacer technique on the ward, every day, to the parent and the child.

Discharge plan

Ready for discharge when

- Stable on salbutamol **3–4-hourly** (not needing it more often);
- **SpO_2 ≥94% in air** (UCG flowchart: >94%);
- PEF **>60–80%** of best or predicted (if able);
- Eating, drinking and talking normally;
- Parent or child has demonstrated **correct inhaler and spacer technique**.

Send home with (the 7-item pack)

1. **Prednisolone** to complete **3 days** (UCG 2023 for children; up to 5 days).
2. **Salbutamol pMDI + spacer**: 2–4 puffs 4-hourly for 1–2 days, then as needed.

3. **Start or step up a controller**: for example **beclomethasone 100 µg twice daily** via spacer (UCG Step 2 for children: 100–200 µg 12-hourly), or ICS-formoterol in adolescents.
4. **Written asthma action plan** with green, yellow and red zones: what to do when well, when getting worse, and when to come straight to hospital.
5. **Trigger advice**: move cooking smoke out of the house or improve ventilation, no smoking indoors, damp-dust, treat colds early.
6. **Follow-up** in 2 days to 1 week, and at 1–3 months.
7. **School letter** allowing the inhaler at school and during sports.

Inhaler and spacer technique (teach, then watch the parent or child do it)

1. Shake the inhaler and remove the cap.
2. Attach it to the spacer.
3. Seal the lips around the mouthpiece; under 3–4 years, hold a face mask over the nose and mouth.
4. Fire **one puff** only.
5. Take **5 tidal breaths** (older children: one slow deep breath and hold for 10 seconds).
6. Wait 30 seconds, shake, and repeat for each puff.
7. Rinse the mouth after an ICS. Wash the spacer weekly and let it air-dry (reduces static).

Bottle spacer: a 500 mL plastic water bottle with a hole cut in the base to fit the inhaler mouthpiece, sealed with tape. It is free and works as well as a commercial spacer.

Counselling: reliever versus preventer. "The blue inhaler opens the airway now. The brown or orange inhaler calms the swelling over weeks: take it every day, even when well." Explain that salbutamol tablets are not a substitute.

Chronic control

GINA 2025 stepwise treatment (current international standard)

Step	Children ≤5 y	Children 6–11 y	Adolescents ≥12 y (Track 1, preferred)
1	As-needed SABA (consider a short course of ICS at the start of a viral illness)	Low-dose ICS taken **whenever SABA is taken**	**As-needed low-dose ICS-formoterol** (anti-inflammatory reliever, AIR)
2	**Daily low-dose ICS** (preferred)	**Daily low-dose ICS**	As-needed low-dose ICS-formoterol
3	**Double** low-dose ICS	Low-dose ICS-LABA, or medium-dose ICS, or very-low-dose ICS-formoterol maintenance and reliever (**MART**)	**Low-dose MART**
4	Continue controller and **refer**	Medium-dose ICS-LABA or low-dose MART; refer	Medium-dose MART
5	n/a	Phenotype; add-on (for example tiotropium, biologics); specialist	Add-on LAMA; phenotype; biologics; specialist

Low daily ICS doses (GINA 2025): beclomethasone dipropionate (HFA, extrafine) **50–100 µg/day** (6–11 y), 100 µg/day (≤5 y); budesonide DPI 100–200 µg/day; fluticasone propionate 50–100 µg/day. **Always via a spacer in children.**

GINA 2025 no longer recommends **SABA-only** treatment for anyone ≥6 years, because it increases the risk of severe attacks.

UCG 2023 chronic asthma steps (older format) Verify *step doses, tablet doses and rescue courses against UCG 2023 5.1.1.2*

UCG step	Treatment (children)
Step 1: intermittent	Salbutamol as needed
Step 2: mild persistent	**Beclomethasone 100–200 µg every 12 hours** + salbutamol as needed
Step 3: moderate persistent	Beclomethasone 100–400 µg 12-hourly (5–12 y); **children <5 years: refer to a specialist**
Step 4: severe persistent	**Refer** (especially children <12 years)
No inhaler available	Salbutamol tablets as a last resort: 2–5 y 1–2 mg per dose; <2 y 100 µg/kg
Rescue prednisolone course	<1 y 1–2 mg/kg; 1–5 y up to 20 mg; 5–15 y up to 40 mg, daily for 3–5 days

Uganda reality: ICS-formoterol (budesonide-formoterol) is available in Kampala private pharmacies and some referral clinics but is expensive; **beclomethasone + salbutamol inhalers are what most Ugandan families can get.**

Every review: check **symptom control, inhaler technique, adherence, triggers and growth** (low-dose ICS has a tiny effect on height, outweighed by the benefit), and step down after 3 months of good control.

10. Complications

Disease-related

Complication	Early recognition	Action
Respiratory failure, arrest	Exhaustion, silent chest, drowsiness, rising $PaCO_2$	ICU, ventilation
Pneumothorax, pneumomediastinum	Sudden deterioration, unilateral reduced air entry, **subcutaneous emphysema**	Chest X-ray; needle decompression then chest drain if tension
Atelectasis, lobar collapse (mucus plug)	Focal reduced breath sounds; collapse on chest X-ray	Usually resolves; physiotherapy after the acute phase
Secondary pneumonia	Fever, focal signs	Antibiotics
Dehydration	Poor intake, fast breathing	Oral or IV maintenance fluids
Long-term	Missed school, poor growth (uncontrolled asthma or repeated oral steroids), fixed airflow limitation, psychosocial stress, death	Controller therapy, education

Treatment-related

Complication	Cause	Recognition and action
Tachycardia, tremor, low K^+, high lactate, high glucose	Salbutamol	Check K^+; judge the asthma by work of breathing and SpO_2, not by HR
Vomiting, arrhythmia, seizures	Aminophylline	Cardiac monitor; stop the infusion
Hypotension, loss of reflexes	Magnesium	Monitor BP and reflexes
Fluid overload, pulmonary oedema	Excess IV fluid	Maintenance only
Respiratory arrest	Sedatives	Never sedate

The salbutamol paradox. A child who has had hourly nebulised salbutamol can be tachycardic, tremulous and anxious, with a high lactate and a low potassium. These are drug effects, not necessarily a worse attack. Look at the work of breathing and SpO_2 to judge the asthma, and check K^+ if salbutamol has been continuous.

11. Prognosis

- Most attacks resolve within days with prompt treatment.
- Deaths are largely preventable: they follow under-grading, too little salbutamol, late oxygen, sedation and discharge without a plan.
- Episodic viral wheeze often settles by school age; atopic, multiple-trigger wheeze is more likely to persist.
- Uncontrolled asthma and repeated oral steroid courses impair growth; low-dose ICS has a very small effect on height, outweighed by benefit.
- Ugandan outcome data are not available.

12. Prevention

- **Primary**: no tobacco smoke in pregnancy or childhood; cook outside or ventilate the kitchen.
- **Secondary**: recognise recurrent and multiple-trigger wheeze early; start a controller; teach spacer technique; give a written action plan and a school letter.
- **Tertiary**: **every admission is a failure of prevention.** After every attack, ask why this child got this bad (no ICS, no spacer, charcoal stove, cost, wrong diagnosis) and fix it. Refer children at Step 4, under-5s needing more than Step 2, any life-threatening attack, poor growth, or doubt about the diagnosis.

13. Special Populations

- **Infants under 2 years**: a first wheeze is usually bronchiolitis; salbutamol response is often poor. Use a spacer with a face mask (under 3–4 years).
- **Adolescents ≥12 years**: ipratropium 500 µg; GINA prefers as-needed ICS-formoterol (Track 1); adherence and smoking need direct questions.
- **HIV**: think LIP, PCP and TB. Ritonavir raises inhaled fluticasone and budesonide levels (Cushing syndrome risk); beclomethasone is usually preferred Verify. Antiretrovirals can alter theophylline levels. Continue cotrimoxazole and ART.
- **On rifampicin**: lower prednisolone and theophylline levels; a higher steroid dose may be needed Verify.
- **Severe acute malnutrition**: potassium is often already low, and salbutamol lowers it further. Check glucose. Follow the SAM fluid protocol.

14. Key Points

High-yield points

- Grade in two minutes: talk or feed, SpO_2, RR, pulse, chest sounds, conscious level.
- **A quiet chest is not a better chest.**
- Oxygen to 94–98%, salbutamol every 20 minutes, steroid within the first hour.
- Not responding: IV magnesium 40–50 mg/kg (max 2 g) over 20 minutes.
- Never sedate, never routine antibiotics, never a β-blocker.
- A "normal" $PaCO_2$ in a breathless child means tiring.
- Salbutamol side effects mimic a worse attack.

- Wheeze with urticaria and hypotension is anaphylaxis: IM adrenaline first.
- A 500 mL bottle spacer works.
- No discharge without a controller, spacer, checked technique, written plan and review date.

Numbers & doses

Item	Value
Severe: HR	**>140 (2–5 y), >125 (>5 y)**
Severe: RR	**>40 (2–5 y), >30 (>5 y)**
Severe: SpO_2 / PEF	**<92%** / **33–50%**
Life-threatening PEF	**<33%**
Oxygen target	**SpO_2 94–98%** (GINA 2025; UCG ≥94%)
Salbutamol pMDI	**2–10 puffs** (100 µg) via spacer every 20 min in the first hour
Salbutamol nebulised	**2.5 mg (<5 y)**; 2.5–5 mg (≥5 y) every 20 min
Ipratropium	**250 µg** nebulised (<12 y), every 20–30 min for the first 2 h
Prednisolone	**1–2 mg/kg** once daily, max **20/30/40 mg** (<2 / 2–5 / ≥6 y), **3 days**
Hydrocortisone	**4 mg/kg IV** 6-hourly, max **100 mg**
IV magnesium sulfate	**40–50 mg/kg over 20 min, max 2 g**
Aminophylline	**5 mg/kg IV over 20 min**, then **1 mg/kg/h**; no load if on theophylline
Adrenaline (anaphylaxis)	**0.01 mg/kg IM** (0.01 mL/kg of 1:1000), max 0.5 mg
Weight estimate (1–10 y)	**(age + 4) × 2 kg**
Discharge	Salbutamol **3–4-hourly**, SpO_2 **≥94% in air**, PEF **>60–80%**, technique checked
Reversibility (spirometry)	FEV_1 rise **≥12%** after bronchodilator
Low-dose ICS (6–11 y)	Beclomethasone **50–100 µg/day** (GINA 2025); UCG 100–200 µg 12-hourly

Exam traps

- **"The wheeze has gone, so he's better."** Not if he is drowsy with a falling RR: silent chest.
- **"Give diazepam because the child is agitated."** Never. Agitation is hypoxia.
- **"Start amoxicillin for every admitted asthmatic."** No: antibiotics only with evidence of bacterial infection.
- **"A nebuliser is always better than a spacer."** No: a spacer is as effective in mild–moderate attacks and teaches technique.
- **"Aminophylline loading dose for everyone."** No: omit the load if the child takes oral theophylline or aminophylline, and magnesium comes first in current guidelines.
- **"Prednisolone must be tapered."** Not after a 3–5-day course.
- **"First wheeze at 6 months = asthma."** Usually bronchiolitis.
- **"A normal $PaCO_2$ is reassuring."** In a breathless asthmatic it means tiring.
- **"Borrow magnesium from maternity and give the eclampsia dose."** The 4 g loading dose is an overdose in a child: 40–50 mg/kg, max 2 g, diluted.
- **"He's on treatment at home."** Salbutamol tablets are not inhaled treatment.

Questions seniors ask

Viva questions

1. *Define asthma.* A chronic inflammatory airway disease with variable symptoms of wheeze, breathlessness, chest tightness and cough, and variable expiratory airflow limitation (GINA 2025).
2. *Name three features of life-threatening asthma.* Silent chest, cyanosis or SpO_2 <92%, poor respiratory effort, exhaustion, altered consciousness, hypotension, PEF <33%.
3. *Salbutamol dose by spacer and by nebuliser?* 2–10 puffs (100 µg each) via spacer; 2.5 mg (<5 y) or 2.5–5 mg (≥5 y) nebulised; every 20 minutes in the first hour.
4. *When do you give ipratropium?* Severe or life-threatening attacks, 250 µg every 20–30 minutes for the first 2 hours, with salbutamol.
5. *Side effects of salbutamol?* Tachycardia, tremor, hypokalaemia, lactic acidosis, hyperglycaemia.
6. *Why is aminophylline less favoured?* Narrow therapeutic index (arrhythmias, seizures, vomiting), interactions, little added benefit over inhaled therapy plus steroids.
7. *What makes a child at high risk of dying from asthma?* Previous ICU or intubation, frequent admissions, SABA overuse, no ICS, poor adherence, psychosocial problems.

On the ward round

- **What is your first action for a child with severe asthma?** Sit them up, oxygen to SpO_2 94–98%, and start salbutamol (with ipratropium if severe) every 20 minutes while you grade the attack and weigh the child.
- **When would you give magnesium, and how much?** For severe or life-threatening asthma not responding to an hour of inhaled salbutamol + ipratropium and a steroid: 40–50 mg/kg IV over 20 minutes, max 2 g, diluted, with BP monitoring.
- **How do you know your treatment is working?** Re-grade every 20–30 minutes: talks in sentences, falling RR and work of breathing, rising SpO_2 and PEF. Tachycardia alone can be a salbutamol effect.
- **What would you send this child home with?** Prednisolone to complete 3 days, salbutamol with a spacer, an inhaled steroid, a written action plan, trigger advice, a school letter and a review in 2–7 days.
- **Why does this child keep coming back?** Usually no controller, poor technique (no spacer), poor adherence, indoor smoke, or the wrong diagnosis (TB, foreign body, heart disease). Check all five.
- **How would you treat anaphylaxis presenting with wheeze?** IM adrenaline 0.01 mg/kg (max 0.5 mg) into the thigh, repeat in 5 minutes if needed, oxygen, fluids for shock, then salbutamol, steroid and antihistamine as add-ons.
- **Which children with asthma need a specialist?** Under-5s needing more than Step 2 (UCG), any child at Step 4, life-threatening attacks, poor growth, doubt about the diagnosis.

Mnemonics

- **Life-threatening asthma, "33, 92, CHEST"**: PEF **<33%**, SpO_2 **<92%**, **C**yanosis, **H**ypotension, **E**xhaustion or altered consciousness, **S**ilent chest, **T**ired (poor effort).
- **Treatment ladder, "O SHIMA"**: **O**xygen, **S**albutamol, **H**ydrocortisone or prednisolone, **I**pratropium, **M**agnesium, **A**minophylline (then ICU).
- **Discharge, "S-P-A-C-E-R"**: **S**teroid course, **P**reventer inhaler, **A**ction plan, **C**heck technique, **E**nvironment and triggers, **R**eview date.
- **Salbutamol side effects, "TTT-L"**: **T**achycardia, **T**remor, **T**oo little potassium, **L**actate.

OSCE tips

- **Inhaler and spacer station**: follow the 7 steps in section 9 (Discharge plan), talking the parent through each. Examiners look for: one puff at a time, 5 tidal breaths (or one deep breath held 10 seconds), a face mask under 3–4 years, rinse after ICS, weekly washing and air-drying, and asking the parent to demonstrate back.
- **PEF station**: stand, reset the needle, deep breath, lips sealed, short sharp blow, best of three, compare with the chart for height.

- **Counselling station**: explain reliever versus preventer ("blue opens the airway now; brown or orange calms the swelling over weeks, take it every day even when well"), the action-plan zones, and smoke avoidance.
- **Examination station**: count RR for a full minute, state SpO_2 thresholds, and present with a severity grade and an immediate plan.

Presenting this tutorial

Edith presents: Week 1, Tuesday, 08:00–09:30, Dr Mawanda.

Goal: in about **40 minutes of presenting + 30 minutes of discussion**, leave every classmate able to **grade an attack, treat it for the first hour, and write a discharge plan**. Dr Mawanda will judge clarity, accuracy of doses, and how you handle questions.

Timed plan

Time	Segment	What to say or do
0–3 min	**Hook case**	Read a 4-line vignette (Brian, section 8: 2 a.m., speaking in words, SpO_2 89%). Ask: "What grade is this, and what is your first action?" Take 2 answers. Do not correct yet
3–8 min	**Definition and burden**	GINA 2025 definition (variable symptoms + variable airflow limitation); exacerbation definition; one Uganda slide: triggers (charcoal smoke, viral colds), what is available at HC III versus RRH
8–14 min	**Pathophysiology in one diagram**	Draw the airway cross-section: smooth muscle (→ salbutamol, ipratropium, magnesium), oedema and inflammation (→ steroids), mucus. Show the $PaCO_2$ phases (low → normal → high)
14–22 min	**Severity grading table**	The single most important slide. Put the UCG 2023 table up and walk the columns. Emphasise **silent chest** and **normal $PaCO_2$ = danger**
22–32 min	**First-hour management algorithm**	Oxygen 94–98% → salbutamol every 20 min (spacer and nebuliser doses) → steroid in the first hour → ipratropium if severe → IV magnesium 40–50 mg/kg, max 2 g → aminophylline → ICU. Then the **drug table** with mg/kg and maxima. Show UCG versus GINA differences in one Uganda box
32–36 min	**Discharge and chronic control**	Discharge criteria, the 7-item discharge pack, GINA 2025 steps in one simplified table, the bottle spacer (bring one and **demonstrate** it)
36–40 min	**Back to the hook case**	Answer it fully; summarise 5 take-home messages
40–70 min	**Questions and discussion**	Use "Questions seniors ask" as backup; ask the class two MCQs from the study set

Tips for presenting well

- **Open with the case and close with the same case.** It makes the talk feel complete.
- **One idea per slide or board section**; no paragraphs. Tables beat text. If there is no projector, draw the **severity table** and the **algorithm** on the board before 08:00.
- **Say doses out loud with units and maxima** ("prednisolone one to two milligrams per kilogram, maximum forty milligrams in a seven-year-old"). Tutors notice dose errors more than anything else.
- **Name your sources**: "UCG 2023 says… GINA 2025 adds…". Have the UCG page numbers ready (acute asthma pp. 331–336, chronic asthma pp. 337–339) Verify *page numbers in the printed UCG 2023*.
- **Demonstrate** the spacer technique (5 breaths per puff, one puff at a time). A 30-second demonstration is remembered better than any slide.
- **Prepare a one-page handout**: the severity table, the first-hour algorithm, and the drug table.

- **If you do not know an answer**, say: "I'm not sure; I'll check the UCG and get back to you after the ward round." Then do it.
- **Rehearse once with a timer** the night before; most first-time presenters overrun by 15 minutes.

Questions Dr Mawanda is likely to ask

1. *How do you distinguish asthma from bronchiolitis in a 14-month-old?* Age, first versus recurrent episode, crackles versus wheeze, atopy, response to salbutamol.
2. *Why is a silent chest dangerous?* No airflow; impending respiratory arrest.
3. *What is the dose and route of magnesium, and why dilute it?* 40–50 mg/kg IV over 20 minutes, max 2 g; the 50% solution is hyperosmolar and irritant, so dilute to ≤20%.
4. *Why don't we give antibiotics?* Exacerbations are mostly viral; antibiotics only for clear bacterial infection.
5. *What would you do differently at an HC III?* Spacer salbutamol, oral prednisolone, observe for 1 hour, refer severe cases with oxygen if available.
6. *How does GINA 2025 differ from UCG 2023 for chronic asthma?* GINA: no SABA-only treatment from 6 years; ICS whenever SABA is used (6–11 y, Step 1); ICS-formoterol as reliever from 12 years. UCG still uses SABA-only Step 1 and beclomethasone at Step 2.

15. References

1. Ministry of Health Uganda. Uganda Clinical Guidelines 2023: National Guidelines for Management of Common Health Conditions. Kampala: MoH; 2023.
2. Ministry of Health Uganda. Essential Medicines and Health Supplies List for Uganda (EMHSLU). Kampala: MoH Verify *edition and year*.
3. Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2025 update.
4. British Thoracic Society, Scottish Intercollegiate Guidelines Network. British guideline on the management of asthma (SIGN 158). 2019.
5. NICE, BTS, SIGN. Asthma: diagnosis, monitoring and chronic asthma management (NG245). 2024.
6. World Health Organization. Pocket Book of Hospital Care for Children, 2nd edition. Geneva: WHO; 2013.
7. Royal Children's Hospital Melbourne. Clinical Practice Guideline: Acute asthma.
8. World Health Organization. Asthma fact sheet.
9. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define asthma (GINA 2025).	A chronic inflammatory airway disease with **variable respiratory symptoms** (wheeze, breathlessness, chest tightness, cough) and **variable expiratory airflow limitation**.
2	Features of a SEVERE asthma attack in a child (UCG 2023/BTS).	Can't complete sentences or too breathless to talk/feed; **SpO_2 <92%**; **HR >140 (2–5 y) or >125 (>5 y)**; **RR >40 (2–5 y) or >30 (>5 y)**; accessory muscle use; **PEF 33–50%**.
3	Features of LIFE-THREATENING asthma.	**Silent chest**, cyanosis, SpO_2 <92%, poor respiratory effort, **exhaustion**, **drowsiness/confusion**, bradycardia or hypotension, **PEF <33%**, normal or rising pCO_2.
4	Oxygen target in acute asthma.	**SpO_2 94–98%** (GINA 2025). UCG 2023: high-flow oxygen at least 5 L/min to keep SpO_2 ≥94%. Use oxygen to drive nebulisers in severe attacks.
5	Salbutamol doses in an acute attack (spacer and nebuliser).	pMDI 100 µg/puff: **2–10 puffs via spacer** (one puff at a time, 5 breaths each). Nebulised: **2.5 mg (<5 y)**, **2.5–5 mg (≥5 y)**. Every **20 min for the first hour**, then by response.
6	Ipratropium in acute asthma: who, dose, how often?	Severe or life-threatening attacks only. **250 µg nebulised** (<12 y; 500 µg ≥12 y) mixed with salbutamol, every **20–30 min for the first 2 h**, then 4–6-hourly (UCG 2023).
7	Oral prednisolone for acute asthma: dose, max, duration.	**1–2 mg/kg once daily** (UCG: 1 mg/kg). Max **20 mg (<2 y), 30 mg (2–5 y), 40 mg (6–11 y)**. **3 days** for children (UCG 2023), up to 5. No taper.
8	IV hydrocortisone dose in paediatric asthma.	**4 mg/kg IV every 6 h**, **max 100 mg** per dose (UCG 2023). Use when vomiting, drowsy or life-threatening; switch to oral prednisolone as soon as possible.
9	IV magnesium sulfate for severe asthma: dose and rate.	**40–50 mg/kg IV (max 2 g) over 20 minutes**, single dose, diluted. Monitor BP and reflexes. First IV bronchodilator in GINA 2025/BTS/RCH.
10	Aminophylline in acute asthma: dose and key caution.	Load **5 mg/kg IV over 20 min**, then **1 mg/kg/h**. **Omit the load if already on oral theophylline/aminophylline.** Narrow therapeutic window (vomiting, arrhythmias, seizures); UCG 2023 discourages its use.
11	Why is a silent chest ominous?	Airways are so narrow that almost **no air moves**, so there is no wheeze. With drowsiness and a falling RR it signals **exhaustion and impending respiratory arrest**.
12	What does a 'normal' pCO_2 mean in a breathless asthmatic child?	**Danger.** Early in an attack the child hyperventilates (low pCO_2). Normal or rising pCO_2 (>6 kPa / 45 mmHg) = **tiring/respiratory failure** → HDU/ICU.
13	Side effects of high-dose salbutamol.	**Tachycardia, tremor, hypokalaemia, lactic acidosis**, hyperglycaemia, agitation. Judge the asthma by work of breathing and SpO_2, not by HR alone.
14	Discharge criteria after an asthma attack.	Stable on salbutamol **3–4-hourly**; **SpO_2 ≥94% in air**; PEF **>60–80%** of best; eating and talking normally; **inhaler/spacer technique checked**.

#	Question	Answer
15	The discharge pack for an asthmatic child (S-P-A-C-E-R).	**S**teroid course to finish, **P**reventer (ICS) started/stepped up, written **A**ction plan, **C**heck technique (spacer), **E**nvironment/trigger advice, **R**eview in 2–7 days.
16	How do you make a spacer from a plastic bottle?	Take a clean **500 mL** water bottle, cut a hole in the base the shape of the inhaler mouthpiece, seal the gap with tape; the child breathes through the bottle neck. Works as well as a commercial spacer.
17	Spacer technique for a child.	Shake inhaler, fit to spacer, seal lips on the mouthpiece (face mask for <3–4 y), **one puff at a time**, **5 tidal breaths** per puff (or a slow deep breath and 10 s hold), wait 30 s, repeat. Rinse mouth after ICS.
18	GINA 2025 Step 1–2 for children 6–11 years.	Step 1: **low-dose ICS taken whenever SABA is taken**. Step 2: **daily low-dose ICS** (plus as-needed SABA). SABA-only treatment is no longer recommended for ≥6 y.
19	GINA 2025 preferred reliever for adolescents ≥12 years.	**As-needed low-dose ICS-formoterol** (anti-inflammatory reliever, AIR) at Steps 1–2; **MART** (ICS-formoterol maintenance and reliever) from Step 3.
20	GINA 2025 for children ≤5 years with frequent wheeze.	Step 2: **daily low-dose ICS** (e.g. beclomethasone 100 µg/day via spacer + mask); Step 3: double the low dose; Step 4: refer. Intermittent ICS at the start of viral illness is an alternative.
21	UCG 2023 chronic asthma Step 2 dose for children.	**Beclomethasone 100–200 µg every 12 h** + salbutamol 1–2 puffs prn. Children **<5 y needing Step 3, and any child needing Step 4: refer** to a specialist.
22	Risk factors for a fatal asthma attack.	Previous **ICU/intubation**; ≥2 admissions or ≥3 ER visits in a year; using >1 salbutamol canister/month; **no ICS**; recent oral steroids; poor adherence; psychosocial problems; food allergy.
23	Wheeze + urticaria + hypotension after eating groundnuts. First drug?	**IM adrenaline 0.01 mg/kg** (0.01 mL/kg of 1:1000), max 0.5 mg, outer thigh; repeat after 5 min if needed. This is **anaphylaxis**, not plain asthma.
24	Why never sedate a child with acute asthma?	Restlessness is usually **hypoxia**. Sedatives (diazepam, promethazine) depress respiratory drive and can precipitate **respiratory arrest**.
25	When is a chest X-ray indicated in acute asthma?	Not routinely. Only for **life-threatening** or non-responding attacks, suspected **pneumothorax** or surgical emphysema, **focal signs/fever** (pneumonia), suspected **foreign body**, or diagnostic doubt.
26	Estimate the weight of a 6-year-old with no scale.	**(age + 4) × 2 = (6 + 4) × 2 = 20 kg** (APLS formula for 1–10 y). Weigh as soon as possible; use a Broselow tape if available.
27	How does UCG 2023 differ from GINA 2025 in the acute attack?	UCG: prednisolone **1 mg/kg**, aminophylline as the IV step, **no magnesium** for children. GINA/BTS: prednisolone 1–2 mg/kg (capped), **IV $MgSO_4$ before aminophylline**. Kampala referral wards use $MgSO_4$ first.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 6-year-old girl (20 kg) with known asthma is brought to Mengo emergency at 3 a.m. She is sitting forward and can say only 2–3 words at a time. RR 36/min, HR 142/min, SpO_2 90% in air, widespread wheeze, temperature 37.4 °C. Which grade of attack is this?

- A Mild, because she is still wheezing
- B Moderate, because SpO_2 is above 88%
- C Severe
- D Life-threatening, because RR is above 30
- E Cannot be graded without a peak flow

2. The girl in the previous question (20 kg, severe asthma, SpO_2 90%) has just been placed on oxygen. Which treatment should be given NEXT?

- A Nebulised salbutamol 5 mg with ipratropium 250 µg, repeated every 20 minutes
- B Oral salbutamol syrup 2 mg and observe for 1 hour
- C IV aminophylline 5 mg/kg as the first bronchodilator
- D Amoxicillin 40 mg/kg and paracetamol
- E Diazepam 0.2 mg/kg IV to calm her so she can use the nebuliser

3. A 4-year-old boy weighing 15 kg is being treated for a moderate asthma attack at a HC IV. He has had salbutamol by spacer. He is not vomiting. What is the correct steroid prescription according to UCG 2023/GINA 2025?

- A Prednisolone 60 mg once daily for 7 days, then taper
- B Dexamethasone 0.6 mg/kg IM once only, because oral steroids are ineffective
- C Hydrocortisone 100 mg IV 6-hourly for 5 days
- D Prednisolone 15–30 mg orally once daily for 3 days
- E No steroid, because steroids are only for severe attacks

4. A 10-year-old boy (30 kg) with severe asthma has had oxygen, three salbutamol + ipratropium nebulisers and IV hydrocortisone over the past hour. He is still speaking in words, SpO_2 91% on 6 L/min oxygen, RR 34, widespread wheeze. Magnesium sulfate 50% is available from the maternity ward. What is the MOST appropriate next step?

- A Magnesium sulfate 4 g IV over 15 minutes, the eclampsia loading dose
- B Magnesium sulfate 1.2–1.5 g IV, diluted, over 20 minutes
- C Stop nebulisers because he has had too much salbutamol
- D Promethazine 0.5 mg/kg IM to settle him
- E Discharge on oral prednisolone because the steroid will work in 6 hours

5. An 8-year-old girl has had hourly nebulised salbutamol for 4 hours. She is now drowsy, RR has fallen from 40 to 20/min, the chest is very quiet on auscultation and SpO_2 is 88% on a face mask. Her mother says she is 'calmer now'. What is the correct interpretation?

- A She is improving; reduce salbutamol to 4-hourly
- B She has salbutamol toxicity; stop all bronchodilators
- C She has fallen asleep from exhaustion after a long night, which is reassuring
- D She has developed pneumonia; start ceftriaxone and continue the same plan
- E She has life-threatening asthma with impending respiratory arrest

6. A 3-year-old boy (14 kg) presents with sudden onset of cough and wheeze that started while he was eating groundnuts with his siblings 2 hours ago. He has no previous wheeze and no fever. Breath sounds are reduced over the right lower zone with a localised wheeze. SpO_2 95%. What is the MOST likely diagnosis?

A Inhaled foreign body
B First episode of asthma
C Bronchiolitis
D Lobar pneumonia
E Anaphylaxis to groundnuts

7. A 12-year-old boy (35 kg) with asthma presents with wheeze, generalised urticaria, lip swelling and BP 70/40 mmHg 15 minutes after a bee sting. Which is the FIRST drug to give?

A Nebulised salbutamol 5 mg
B Hydrocortisone 100 mg IV
C Adrenaline 0.35 mg (0.35 mL of 1:1000) IM into the thigh
D Chlorphenamine 4 mg orally
E Aminophylline 5 mg/kg IV

8. A 7-year-old girl was admitted with severe asthma and is now on salbutamol by spacer 4-hourly. SpO_2 is 96% in air, she is eating and running around the ward. Her only home treatment has been salbutamol tablets bought from a drug shop. Which discharge plan is MOST appropriate?

A Continue salbutamol tablets 2 mg three times daily and review if she wheezes again
B Oral prednisolone for 1 month to prevent relapse
C Salbutamol inhaler without a spacer, as needed
D Amoxicillin for 5 days to prevent a chest infection, and salbutamol tablets
E Finish prednisolone, salbutamol inhaler with a spacer, start beclomethasone via spacer, written action plan and review in 1 week

9. A 14-month-old girl presents in the rainy season with 2 days of runny nose, then cough, fast breathing and difficulty feeding. RR 58/min, SpO_2 93%, bilateral fine crackles and some wheeze. This is her first episode. A trial of salbutamol produces no change. Which diagnosis is MOST likely?

A Acute severe asthma
B Bronchiolitis
C Inhaled foreign body
D Heart failure from a VSD
E Pertussis

10. A 9-year-old boy with severe asthma has received IV magnesium. He is started on IV aminophylline. Which history item would make you OMIT the loading dose? Advanced

A He received hydrocortisone 1 hour ago
B He is HIV-positive on dolutegravir-based ART
C He has had three salbutamol nebulisers
D He has been taking oral aminophylline tablets at home
E He is allergic to penicillin

11. After 3 hours of frequent nebulised salbutamol, a 6-year-old with improving asthma has HR 165/min, a fine tremor, and serum potassium 2.9 mmol/L. She now speaks in sentences and SpO_2 is 96% on 1 L/min. What is the BEST explanation and response? Advanced

- A Worsening asthma; give IV aminophylline
- B Sepsis; take blood cultures and give ceftriaxone
- C Expected β_2-agonist effects; space salbutamol out and correct potassium
- D Supraventricular tachycardia; give adenosine
- E Anxiety; give oral diazepam

12. A 10-year-old girl with asthma uses her salbutamol inhaler 4–5 times a week and wakes at night coughing twice a week. She has no inhaled steroid. According to GINA 2025, what is the preferred controller?

- A Daily low-dose inhaled corticosteroid via a spacer
- B Salbutamol tablets 2 mg three times daily
- C Oral prednisolone 10 mg daily
- D Montelukast alone as first choice
- E Increase salbutamol to regular 4-hourly use

13. Which ONE of the following is a recognised feature of life-threatening (rather than severe) asthma in a 7-year-old?

- A Heart rate 130/min
- B Respiratory rate 34/min
- C Use of accessory muscles
- D Altered level of consciousness
- E Peak flow 45% of predicted

14. A 5-year-old (18 kg) with severe asthma is vomiting every oral dose. You need to give a systemic steroid. What is the MOST appropriate prescription?

- A Prednisolone 18 mg orally; repeat if vomited
- B Hydrocortisone 72 mg IV every 6 hours until oral prednisolone is tolerated
- C Dexamethasone 16 mg IV every 6 hours
- D Beclomethasone 400 µg inhaled as a single high dose instead
- E Methylprednisolone 30 mg/kg IV daily for 3 days

15. At a HC III with no oxygen, a 9-year-old girl (25 kg) with moderate asthma has received salbutamol 6 puffs via a bottle spacer three times over an hour and prednisolone 25 mg. She now speaks in sentences, RR 24/min, no recession. What should you do?

- A Refer immediately to the regional referral hospital
- B Give aminophylline 5 mg/kg IV at the HC III
- C Admit at the HC III for 5 days of IV hydrocortisone
- D Discharge with salbutamol tablets only
- E Observe for a further 1–3 hours; if stable, discharge with prednisolone to complete 3 days, inhaler + spacer, ICS, action plan and review in 48 hours

Answers & explanations

1. C · Severe

C is right: inability to complete sentences, SpO_2 <92%, HR >125 and RR >30 in a child >5 years are all **severe** features. A is wrong: wheeze is present in every grade except the silent chest; it does not mean mild. B is wrong: the cut-off is 92%, not 88%. D is wrong: RR >30 is a severe, not life-threatening, feature; she has no silent chest, cyanosis, exhaustion or altered consciousness. E is wrong: grading is clinical; PEF helps but is not required (and is unreliable in a breathless 6-year-old).

2. A · Nebulised salbutamol 5 mg with ipratropium 250 µg, repeated every 20 minutes

A is right: in a severe attack the first-line treatment is inhaled **salbutamol plus ipratropium**, oxygen-driven, every 20 minutes for the first hour (UCG 2023, GINA 2025). B is wrong: oral salbutamol is slow, weak and causes side effects; it is not acute treatment. C is wrong: aminophylline is a later step after inhaled therapy, steroid (and magnesium) have failed. D is wrong: there is no evidence of bacterial infection; antibiotics are not routine. E is wrong: sedation in asthma can cause respiratory arrest; agitation is hypoxia.

3. D · Prednisolone 15–30 mg orally once daily for 3 days

D is right: prednisolone **1–2 mg/kg** (15–30 mg for 15 kg; max 30 mg at 2–5 years) once daily for **3 days** is the paediatric course. A is wrong: 60 mg is far above the maximum for a 4-year-old, 7 days is unnecessarily long, and no taper is needed for short courses. B is wrong: oral steroids work as well as injected ones when the child can swallow. C is wrong: hydrocortisone is for children who cannot take oral medication, and the dose is 4 mg/kg (60 mg), not 100 mg. E is wrong: any attack needing more than a single dose of salbutamol should get a steroid early.

4. B · Magnesium sulfate 1.2–1.5 g IV, diluted, over 20 minutes

B is right: IV $MgSO_4$ **40–50 mg/kg** (30 kg → 1.2–1.5 g; max 2 g), diluted, over 20 minutes is the first IV bronchodilator for severe asthma not responding to an hour of inhaled therapy (GINA 2025, BTS/SIGN, RCH). A is wrong: the 4 g eclampsia load is double the paediatric maximum and could cause hypotension and respiratory depression. C is wrong: he is still severe; inhaled salbutamol should continue (1-hourly or back-to-back). D is wrong: sedation is dangerous in asthma. E is wrong: he is hypoxic and severe; discharge would be unsafe.

5. E · She has life-threatening asthma with impending respiratory arrest

E is right: drowsiness, a falling RR, a **silent chest** and low SpO_2 are classic signs of **exhaustion and impending arrest**: call senior help, back-to-back nebulisers, IV magnesium, prepare to ventilate. A is wrong: a falling RR without better SpO_2 and alertness is deterioration, not improvement. B is wrong: salbutamol toxicity causes tachycardia and tremor, not a silent chest; bronchodilators must continue. C is wrong: altered consciousness in asthma is a life-threatening feature. D is wrong: nothing suggests pneumonia, and antibiotics would not address the immediate threat.

6. A · Inhaled foreign body

A is right: **sudden onset while eating**, no previous wheeze, and **unilateral** reduced air entry with localised wheeze = **foreign body** (usually right main bronchus) until proven otherwise; get inspiratory and expiratory CXR and refer for bronchoscopy. B is wrong: asthma gives bilateral, diffuse signs and usually a history of recurrent episodes. C is wrong: bronchiolitis affects infants <2 years after a coryzal prodrome, with bilateral crackles/wheeze. D is wrong: there is no fever and the onset was sudden. E is wrong: anaphylaxis would have urticaria, angioedema, vomiting or hypotension and bilateral wheeze.

7. C · Adrenaline 0.35 mg (0.35 mL of 1:1000) IM into the thigh

C is right: this is **anaphylaxis** (skin + respiratory + hypotension after a sting). First drug: **IM adrenaline 0.01 mg/kg** (35 kg → 0.35 mg; max 0.5 mg) into the outer thigh, repeated after 5 minutes if needed. A is wrong: salbutamol helps bronchospasm but does nothing for the shock and angioedema; it is an add-on. B is wrong: steroids act over hours and are adjuncts. D is wrong: antihistamines treat itch only and never replace adrenaline. E is wrong: aminophylline has no role in anaphylaxis.

8. E · Finish prednisolone, salbutamol inhaler with a spacer, start beclomethasone via spacer, written action plan and review in 1 week

E is right: she meets discharge criteria; the plan must **finish the steroid course**, provide an inhaled reliever **with a spacer**, **start a controller** (ICS; she has had a severe attack, so Step 2 or above), a written action plan, trigger advice and early follow-up. A is wrong: tablets are slow and side-effect-prone and leave the inflammation untreated. B is wrong: long courses of oral steroids cause growth suppression and other harm; inhaled steroids are the controller. C is wrong: without a spacer a child gets very little drug, and SABA-only therapy is not recommended for ≥6 y (GINA 2025). D is wrong: prophylactic antibiotics have no role.

9. B · Bronchiolitis

B is right: a **first** episode in a child <2 years after a **coryzal prodrome**, with **fine crackles** and wheeze and **no response to salbutamol**, is bronchiolitis (usually RSV). A is wrong: asthma is rarely diagnosed at this age after a first episode, and crackles are not typical. C is wrong: a foreign body causes sudden onset and unilateral signs. D is wrong: there is no murmur, hepatomegaly or chronic poor feeding/growth history. E is wrong: pertussis causes paroxysmal cough with whoop or apnoea, usually with a clear chest between paroxysms.

10. D · He has been taking oral aminophylline tablets at home

D is right: if a child already takes **oral theophylline/aminophylline**, a loading dose risks **toxic levels** (vomiting, arrhythmias, seizures); start the maintenance infusion only, ideally guided by a level. A is wrong: steroids do not interact with aminophylline in this way. B is wrong: dolutegravir does not mandate omitting the load (rifampicin and some PIs alter theophylline levels, but not DTG). C is wrong: salbutamol does not change the loading decision, though both drugs can lower potassium. E is wrong: penicillin allergy is irrelevant to aminophylline dosing.

11. C · Expected β_2-agonist effects; space salbutamol out and correct potassium

C is right: tachycardia, tremor and **hypokalaemia** are predictable **salbutamol effects**; her asthma is improving (talking, SpO_2 96%), so space doses out and add potassium to fluids. A is wrong: her work of breathing and SpO_2 show improvement, and aminophylline would worsen tachycardia and hypokalaemia. B is wrong: there is no fever or other sign of sepsis. D is wrong: sinus tachycardia of 165 with a clear cause is not SVT (usually >220 in children, fixed rate). E is wrong: sedation is contraindicated in acute asthma.

12. A · Daily low-dose inhaled corticosteroid via a spacer

A is right: symptoms more than twice a week with night waking need a **daily low-dose ICS** (Step 2–3 for 6–11 years) with a spacer; UCG 2023 Step 2 is beclomethasone 100–200 µg 12-hourly. B is wrong: oral β_2-agonists are not controllers. C is wrong: daily oral steroids are reserved for the most severe asthma under specialist care. D is wrong: montelukast is a less effective alternative, not first choice. E is wrong: regular SABA without ICS increases the risk of severe attacks.

13. D · Altered level of consciousness

D is right: **altered consciousness** (drowsiness, confusion) is a life-threatening feature. A is wrong: HR >125 in a child >5 y is a severe feature. B is wrong: RR >30 is severe. C is wrong: accessory muscle use is severe. E is wrong: PEF 33–50% is severe; life-threatening is <33%.

14. B · Hydrocortisone 72 mg IV every 6 hours until oral prednisolone is tolerated

B is right: **hydrocortisone 4 mg/kg IV** (18 kg → 72 mg; max 100 mg) every 6 hours, then oral prednisolone once tolerated (UCG 2023). A is wrong: if she is vomiting every dose, the oral route is unreliable. C is wrong: 16 mg 6-hourly is a large overdose for an 18 kg child. D is wrong: inhaled steroids do not replace systemic steroids in a severe attack. E is wrong: 30 mg/kg is a pulse dose used in conditions such as lupus or transplant rejection, not asthma.

15. E · Observe for a further 1–3 hours; if stable, discharge with prednisolone to complete 3 days, inhaler + spacer, ICS, action plan and review in 48 hours

E is right: she has responded well. UCG 2023 says monitor for 30–60 min (or longer), then send home with prednisolone (3 days in children), start or step up chronic treatment, teach self-management and **review in 48 hours**. A is wrong: referral is for poor response, relapse within 3–4 h, or severe features. B is wrong: aminophylline is not needed in a responding child and is not a HC III drug. C is wrong: she is not severe and can take oral medication. D is wrong: tablets alone ignore both the reliever technique and the controller.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Night-time severe asthma in a 7-year-old at Mengo

Brian, a 7-year-old boy from Kawempe, is carried into Mengo Hospital paediatric emergency at 2 a.m. by his mother. He has had a cold for 2 days and became very breathless tonight. He is sitting forward, can say only single words, and looks frightened. He is known to have asthma and uses a salbutamol inhaler 'when it is bad', without a spacer.

1. What do you do in the first 2 minutes?

> Move him to the resuscitation area, **sit him up** on his mother's lap, start **oxygen** (face mask 5–10 L/min) and attach a pulse oximeter. While a colleague prepares salbutamol, **grade the attack**: talk/feed, SpO_2, RR, HR, chest sounds, consciousness. Weigh him (or estimate (7 + 4) × 2 = 22 kg).

2. Observations: weight 22 kg, SpO_2 89% in air, RR 38, HR 138, T 37.6 °C, widespread wheeze, equal air entry, intercostal recession, alert. PEF 40%. Grade the attack.

> **Acute severe asthma**: speaks in words, SpO_2 <92%, RR >30, HR >125, PEF 33–50%. Not life-threatening: alert, wheeze present (not silent), no cyanosis, PEF >33%.

3. Write the first-hour treatment orders.

> 1. Oxygen to SpO_2 **94–98%**.
> 2. **Salbutamol 5 mg + ipratropium 250 µg** nebulised with oxygen at 6–8 L/min, **every 20 min × 3**.
> 3. **Prednisolone 40 mg** orally now (1–2 mg/kg, max 40 mg at ≥6 y) or hydrocortisone 88 mg IV if vomiting.
> 4. Observations every 15 min; re-grade at 20, 40, 60 min.
> 5. No sedatives, no antibiotics (no focal signs).

4. History taken during the nebulisers: two admissions this year, never in ICU, no inhaled steroid, charcoal stove indoors, father smokes, fully immunised, HIV test negative last year, WAZ −0.5. What does this add?

> He has **uncontrolled asthma** with risk factors for a severe attack (**no ICS**, repeated admissions, smoke exposure). It does not change the immediate plan but drives the discharge plan: controller, spacer, smoke advice, action plan. No ICU history is reassuring but does not lower vigilance.

5. At 60 min: speaks in sentences, SpO_2 95% on 2 L, RR 28, HR 150 with tremor, PEF 65%. Interpret and plan.

> **Good response**; the higher HR and tremor are **salbutamol effects**. Admit for observation: salbutamol (spacer) 1–2-hourly then 4-hourly, ipratropium can stop after 2–6 h, continue prednisolone daily for 3 days, wean oxygen to keep ≥94%, check K^+ if salbutamol stays frequent.

6. Next morning he is on salbutamol 4-hourly, SpO_2 96% in air, eating well. Write his discharge plan.

> **S-P-A-C-E-R**: finish **prednisolone** (2 more days); salbutamol inhaler 2–4 puffs 4-hourly for 1–2 days then as needed, **with a bottle spacer**; start **beclomethasone 100 µg twice daily** via spacer (UCG Step 2); **written action plan**; **environment** (cook outside or ventilate, father not to smoke indoors); technique checked by teach-back; **review in 1 week** and school letter.

Take-away: Grade in 2 minutes, treat by grade every 20 minutes, give the steroid in the first hour, and use the admission to fix control: spacer, inhaled steroid, smoke and an action plan.

Case 2: The quiet chest: life-threatening asthma

Aisha, a 10-year-old girl weighing 30 kg, has been on the paediatric ward for 3 hours with severe asthma. She has had five salbutamol + ipratropium nebulisers and IV hydrocortisone. The night nurse calls you because 'she has calmed down and is sleeping, but the saturation machine keeps alarming'.

1. What do you expect to find, and what are you worried about?

> Worry about **exhaustion**: a child who 'calms down' with falling SpO_2 may be tiring. Look for **drowsiness, a falling RR, a silent chest, cyanosis, bradycardia** and poor effort; these are life-threatening features.

2. Findings: rousable only to voice, RR 18, very quiet chest bilaterally, SpO_2 86% on 10 L mask, HR 150, BP 95/60. What is your diagnosis and first actions?

> **Life-threatening asthma** with impending respiratory arrest. Call the **most senior clinician and anaesthetist** now. High-flow oxygen; **back-to-back salbutamol + ipratropium**; bag-valve-mask ready; second IV line; continue hydrocortisone.

3. Which IV drug would you give next, and at what dose?

> **Magnesium sulfate 40–50 mg/kg IV = 1.2–1.5 g** (max 2 g), diluted to ≤20% (e.g. 3 mL of 50% in 0.9% saline), **over 20 minutes**; monitor BP and reflexes. Borrow from maternity if needed, but never give the 4 g eclampsia dose.

4. A capillary gas shows pH 7.28, pCO_2 6.8 kPa (51 mmHg), lactate 3.8 mmol/L. Interpret.

> **Respiratory acidosis** with a raised pCO_2 in a breathless asthmatic = **respiratory failure** from exhaustion. The lactate is partly from salbutamol. She needs HDU/ICU and probably ventilation.

5. Twenty minutes after magnesium she is still drowsy with pCO_2 rising. What next?

> **Aminophylline 5 mg/kg (150 mg) IV over 20 min** (if not on oral theophylline), then 1 mg/kg/h with cardiac monitoring, or IV salbutamol in HDU. Prepare for **intubation** by the most experienced person (ketamine induction; slow ventilation rate, long expiratory time). Arrange **PICU** (Mulago or private) and transfer with oxygen and an escort if not available on site.

6. She recovers after 2 days in ICU. What must the discharge and follow-up include?

> She is now **high risk for fatal asthma** (ICU admission): specialist asthma clinic, **daily ICS (or ICS-LABA/MART per GINA 2025)**, spacer, written action plan with clear 'go to hospital now' criteria, adherence support, trigger control, and early review (within 1 week).

Take-away: A calm, drowsy asthmatic with a quiet chest and falling SpO_2 is dying, not improving: call for help, magnesium, then aminophylline/ICU.

Case 3: Recurrent wheeze in a toddler at a HC IV

Joel, a 3-year-old boy weighing 12 kg, is brought to a HC IV in Mukono with cough and wheeze for 1 day. His mother says it is the 'fourth time this year'; each time he gets salbutamol syrup. Between episodes he coughs at night and when he runs. His father has hay fever.

1. What are your differentials and what in the history points to asthma?

> Preschool **multiple-trigger wheeze/asthma** (recurrent episodes, night and exercise cough, atopic family) vs **episodic viral wheeze**; also consider TB (contact? weight loss?), foreign body (sudden onset?), HIV-related lung disease, heart failure, recurrent aspiration. Symptoms **between** colds and **atopy** favour asthma.

2. Examination: alert, talking in phrases, RR 38, HR 130, SpO_2 94% in air, mild subcostal recession, bilateral wheeze, no crackles, afebrile. WAZ −1.2, BCG scar present. Grade and treat.

> **Mild–moderate** attack (SpO_2 ≥92%, RR ≤40, HR ≤140, talking). Give **salbutamol 2–6 puffs via spacer with a face mask** (or 2.5 mg nebulised) every 20 min for an hour, and **prednisolone 12–24 mg** (1–2 mg/kg; max 30 mg at 2–5 y) orally. Re-grade after each dose.

3. He improves after 2 rounds. What should you check before deciding on chronic treatment?

> **Growth** (WAZ −1.2: monitor), **TB exposure** and HIV status (offer a test if unknown), **immunisation** card (UNEPI: PCV, measles-rubella), feeding, **home smoke**, and whether the family can afford/use an inhaler with a spacer and mask.

4. What controller does GINA 2025 recommend for him, and what does UCG 2023 say?

> GINA 2025 (≤5 y): frequent episodes (≥3 per year) and interval symptoms → **daily low-dose ICS** (e.g. beclomethasone 100 µg/day via spacer + face mask), reviewed after 2–3 months. UCG 2023: Step 2 beclomethasone 100–200 µg 12-hourly; if a child **<5 y needs Step 3 or more, refer** to a specialist.

5. How do you teach the mother to use the inhaler, and what do you tell her about salbutamol syrup?

> Shake, attach to a spacer with a **face mask**, mask sealed over nose and mouth, **one puff, count 5–6 breaths**, repeat; rinse mouth/wipe face after ICS; clean the spacer weekly. Explain that **syrup is slow and causes shaking and fast heartbeat**; the inhaler with a spacer is quicker, safer and cheaper per dose. Give a written plan and review in 2–4 weeks.

Take-away: Recurrent wheeze with interval symptoms in an atopic toddler deserves an inhaled steroid via a spacer and mask, not repeated salbutamol syrup; check growth, TB and HIV along the way.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.